# Evaluating Prompt Consistency in Financial Language Models

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***Abstract*—Large Language Models (LLMs) are increasingly used to interpret financial disclosures such as earnings calls and analyst commentary, and to form predictions about near-term stock performance. However, traditional metrics such as accuracy or MCC do not capture whether these models behave in a stable and trustworthy way when faced with paraphrases, minor wording changes, or alternative prompting styles. Prior work (e.g., FinTrust) showed that even small semantic-preserving edits—such as negation or symmetric rephrasing—can cause sharp drops in forecast quality, suggesting a lack of consistency that would be unacceptable in real decision-making.**

**In this work we present a targeted and feasible study, FinTrust-2025, focusing on prompt-based consistency for financial LLMs under a short-horizon direction task. We curate a 2024–2025 dataset of $N = 486$ one-sentence statements drawn from 81 earnings calls across 12 large-cap firms in four industries, with ground-truth labels defined by 3-trading-day close-to-close movement after the earnings event. We evaluate a deterministic GPT-4o prompting pipeline (*Trinity*) and stress-test robustness via controlled semantic transformations (Negation / Symmetry / Additive / Transitive). We summarize behavioral stability with a transformation-based Prompt Consistency Index (PCI). Empirically, Trinity attains 0.585 accuracy on the non-neutral subset, and exhibits high invariance under symmetry/transitive/additive transformations (98.0–99.1%) but rarely flips under a lexical negation test (1.0%), highlighting a key gap between "stable" outputs and logically expected behavior under polarity reversal.**

**Our goal is not to propose a better trading model; instead, we deliver an evaluation pipeline that quantifies how fragile or reliable a financial LLM is under realistic variations in wording and prompting. This establishes an actionable benchmark for future extensions such as improved negation handling, richer transformation validity checks, and longer-horizon evaluation.**

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## I. Introduction

Financial NLP has moved rapidly from sentiment lexicons toward neural forecasters that ingest full earnings-call transcripts and managerial guidance. Analysts already ask LLMs questions such as "Will the outlook next quarter be positive or negative?" and "Is management signaling cost pressure or margin expansion?" The appeal is obvious: models can summarize narratives, extract tone, and suggest directional views in seconds.

But there is a basic trust problem. A model that looks "accurate" on average may still be operationally dangerous if its answer changes drastically when the wording of the question changes slightly. Recent work (FinTrust) showed that even when two inputs are logically equivalent—or differ only via controlled transformations such as **Negation**, **Symmetry**, **Additive** combination, or **Transitive** inference—the predicted movement label and even the simulated trading outcome can flip unpredictably. This exposes a gap that classic accuracy metrics fail to measure: **consistency under meaning-preserving variation**.

At the same time, current evaluations face three limitations:

1) They rarely test the **prompt sensitivity** of modern LLMs. In practice, different users pose economically identical questions in very different styles (e.g., "You are a risk officer, assess downside" vs. "Will the stock go up? Answer UP or DOWN"). If an LLM gives incompatible answers across prompts, it is not deployable.
2) They are typically built on older time periods and do not reflect more recent disclosures and vocabulary from 2024–2025.
3) Their metrics are not explicitly designed to separate decision stability ("Do you still say UP vs. DOWN?") from reasoning stability ("Do you still justify it the same way?").

In this project we present a focused and time-feasible study called **FinTrust-2025** that targets these issues in a short-horizon setting. Concretely, we:

- Construct a recent multi-industry dataset of earnings-call statements (2024–2025), with a precise 3-trading-day labeling protocol.
- Evaluate a deterministic GPT-4o prompting pipeline (*Trinity*) that outputs {`UP`, `DOWN`, `NEUTRAL`} under structured prompting and explicit abstention.
- Stress-test behavioral stability via FinTrust-style transformations (Negation / Symmetry / Additive / Transitive) and quantify stability using a transformation-based Prompt Consistency Index (PCI).

Importantly, we deliberately scope our contribution to **prompt-based consistency** and transformation-based stress testing. More ambitious elements—such as richer causal consistency checks, risk-aware indices, and end-to-end backtesting tied to realized PnL—are left for future work; here we focus on delivering a reusable benchmark and evaluation pipeline for consistency in financial LLMs.

## II. Related Work

**Text-based financial forecasting.** Prior work has shown that managerial language during earnings calls carries predic-

tive signals for future market behavior. For example, verbal content and vocal delivery (tone, stress) from executives have been used to forecast subsequent stock volatility, suggesting that both "what you say" and "how you say it" matter for financial risk assessment [1]. Later work moved to deep architectures: hierarchical Transformer models were trained in a multi-task setup to model price/volatility dynamics at multiple textual granularities (utterance → section → full call), improving predictive power on financial targets [2]. Domain-specific language models such as FinBERT further adapted general-purpose Transformers to finance, showing that pre-training on financial text can boost downstream sentiment and movement classification performance [3]. More recent systems directly treat earnings-call disclosures as inputs and attempt to infer short-horizon stock movement or directional sentiment for decision support [4]. However, most approaches are still evaluated mainly with standard metrics such as accuracy or regression error and do not explicitly ask whether the model is behaviorally stable under small but meaningful variations.

**Behavioral robustness and consistency testing.** In general NLP, researchers have argued that raw accuracy is not sufficient to judge reliability. CheckList formalized "behavioral testing": instead of only scoring correctness, it probes models with targeted capability tests and controlled perturbations to reveal brittle behaviors that normal evaluation hides [5]. Follow-up work showed that even when two inputs are logically equivalent or differ only through minor paraphrasing, models may produce contradictory answers; this motivates evaluation criteria based on invariance under meaning-preserving edits, not just task accuracy [6]. In parallel, prompting research has demonstrated that eliciting intermediate reasoning steps can improve reliability on complex tasks; Chain-of-Thought (CoT) prompting is a canonical approach to induce step-by-step reasoning in LLMs [7]. We leverage this principle in our prompting design by enforcing a phase-wise protocol with explicit intermediate fields.

**FinTrust-style financial consistency.** FinTrust brought behavioral robustness into financial forecasting by defining four structured "consistency transformations" for financial text—Negation, Symmetry, Additive, and Transitive—and using them to stress-test finance prediction models [4]. It reported that seemingly small, semantics-preserving edits can cause severe drops in classification performance and can even flip simulated trading outcomes. Importantly, it also highlighted two open issues: (i) consistency and accuracy can be weakly correlated, meaning a model can look "good" while being logically unstable; and (ii) LLMs were not yet systematically evaluated for *prompt sensitivity*, i.e., whether they change their forecast simply because the user phrases the same economic question in a different way [4].

**Pipeline decomposition and financial LLMs.** Beyond finance, ChartGPT shows that LLM outputs can become more controllable when generation is decomposed into an explicit multi-step pipeline with structured intermediate outputs that can be inspected [8]. This design philosophy aligns with our Trinity prompting pipeline, which decomposes prediction into extraction, debate, and verdict stages under a strict JSON schema. Meanwhile, FinGPT advocates an open-source and data-centric approach to building finance-specialized LLMs via automatic data curation and lightweight adaptation (e.g., LoRA), supporting a broad range of financial NLP applications [9]. Our work is complementary: rather than proposing a new finetuned financial LLM, we focus on evaluating and stress-testing a deterministic prompting pipeline using transformation-based consistency metrics.

| Sector | #Calls | #Stmts | UP | DOWN |
|---|---|---|---|---|
| Technology | 21 | 126 | 54 | 72 |
| Financials | 19 | 114 | 66 | 48 |
| Retail | 23 | 138 | 66 | 72 |
| Healthcare | 18 | 108 | 66 | 42 |
| **Total** | 81 | 486 | 252 | 234 |

TABLE I
DATASET COMPOSITION AND GROUND-TRUTH LABELS.

**Our focus.** We target the remaining gap identified above. We do not propose yet another forecasting architecture; instead, we contribute an evaluation protocol for **prompt-based consistency in financial LLMs** using recent (2024–2025) earnings-call text. Concretely, we (i) ask the same economic question through multiple prompt styles (different analyst roles, output formats, and horizon phrasings), (ii) apply FinTrust-style semantic transformations to the underlying statement, and (iii) quantify whether the model's directional decision (and a short justification) stays stable. We summarize this stability with a single score, the Prompt Consistency Index (PCI), with the goal of making "consistency as reliability" measurable for financial LLMs.

## III. Dataset and Labeling Protocol

### A. Company coverage and sampling

We manually selected 12 companies across four industries (3 companies per industry). Technology: AAPL (2024Q1–2025Q4), PYPL (2024Q1–2024Q4, 2025Q2–2025Q3), NVDA (2024Q1–2024Q3, 2025Q1–2025Q4). Financials: JPM (2024Q1–2025Q3), C (2024Q1–2025Q3), BLK (2024Q1–2024Q4, 2025Q2). Retail: WMT (2024Q1–2025Q4), COST (2024Q1–2025Q4), KO (2024Q1–2025Q3). Healthcare: JNJ (2024Q1–2025Q3), LLY (2024Q1–2024Q4, 2025Q3), ISRG (2024Q1–2024Q4, 2025Q2–2025Q3).

For each earnings call (company-quarter), we extracted exactly 6 short statements. Each statement is approximately one sentence (at most one period; if a comma appears, the statement can contain two short clauses). This yields 81 calls $\times$ 6 statements/call $= 486$ statement-level samples.

### B. Ground-truth label (3 trading days)

We label each statement using the stock **close-to-close** movement over **3 trading days**. Let $t_0$ be the earnings event

day; if the call occurs outside regular market hours, $t_0$ is set to the next trading day. We define:

$$\text{label} = \begin{cases} \texttt{UP}, & \text{if Close}(t_0{+}3) > \text{Close}(t_0), \\ \texttt{DOWN}, & \text{otherwise.} \end{cases}$$

Trading days follow the market calendar (skipping weekends/holidays). We use the same price series consistently within each ticker (e.g., close or adjusted close) to avoid corporate-action artifacts. **Note.** We only report 3-day results in this report; 7/15/30-day horizons are left for future work.

### C. Auxiliary attributes: semantic class and tone

Each sample includes two auxiliary attributes: `semantic_class` (e.g., *Growth*) and `tone` (*Positive/Neutral/Negative*). These attributes are assigned via a GPT-based classifier by providing a JSON schema example and instructing the model to fill the corresponding fields. We constrain the output space to the three tone labels and ask the model to flag mixed-tone cases for manual review.

## IV. Method

### A. Overview

We evaluate a deterministic LLM-based pipeline (*Trinity*) for short-horizon post-earnings direction prediction and quantify robustness via transformation-based consistency. Given a statement-level sample with its company-quarter context, we (i) construct a high-signal hybrid context, (ii) query GPT-4o with a structured protocol to obtain a direction in {`UP`, `DOWN`, `NEUTRAL`}, and (iii) apply controlled statement transformations (Negation / Symmetry / Transitive / Additive).

We summarize consistency using two complementary measures: **Decision Consistency Rate (DCR)** is our strict model-comparison score computed only on *non-abstained* pairs (both original and transformed decisions are in {`UP`,`DOWN`}). **Prompt Consistency Index (PCI)** is a quarter-aggregated diagnostic score used for visualization; it includes `NEUTRAL` by mapping `UP`/`DOWN`/`NEUTRAL` to $+1/-1/0$ and averaging consistency scores within each quarter.

### B. Trinity prompting pipeline

*a) The Trinity protocol (summary).:* We implement a deterministic three-part analysis pipeline:

1) **Hybrid forensic retrieval (RAG-inspired context selection)** [10]: we retain a tone-preserving *anchor* (first 2500 characters) and select Top-$K$ *evidence* sentences via weighted keyword scoring (guidance keywords prioritized over other signals).
2) **Structured reasoning (Extraction → Debate → Verdict)** [7]: the model must fill phase-wise intermediate fields before producing the final decision under a strict JSON schema.
3) **Python guardrails (hard rules)**: if a guidance cut or momentum deceleration is detected from extracted fields, the final decision is deterministically overridden to `DOWN`.

*b) Prompt protocol and output schema.:* Trinity enforces a three-phase protocol: *(1) Extraction* of discrete signals (`guidance_trend`, `revenue_performance`, `momentum`), *(2) Debate* among `Bull/Bear/Skeptic` perspectives, and *(3) Verdict* mapping extracted signals to a final directional decision. The model is required to output a JSON object with the following fields:

```
{
  "Phase1_Extraction": {
    "guidance_trend": "...",
    "revenue_performance": "...",
    "momentum": "..."
  },
  "Phase2_Debate_Summary": "...",
  "Phase3_Verdict": {
    "final_decision": "Up" or "Down" or
    ↪  "Neutral",
    "confidence": "High" or "Medium" or "Low"
  }
}
```

We map `final_decision` to {`UP`, `DOWN`, `NEUTRAL`}, where `NEUTRAL` indicates abstention (low-confidence output).

*1) Hybrid forensic retrieval (context construction):* To reduce context dilution, we construct a hybrid context from the company-quarter call-level text formed by concatenating the 6 extracted statements: (i) a **tone anchor** consisting of the first 2500 characters, and (ii) **sniper evidence** consisting of Top-$K$ sentences selected from the remaining text by a weighted keyword scoring function. We prioritize guidance-related signals by assigning the highest weights to guidance keywords, with additional emphasis on negative guidance updates (e.g., "lower" / "cut"). The final hybrid context concatenates the tone anchor and the Top-$K$ evidence sentences. This context-selection step is *retrieval-style* (within-call evidence selection) and is conceptually inspired by retrieval-augmented generation pipelines [10].

*2) Neuro-symbolic override rules (Python guardrails):* We add deterministic Python-side hard rules as a safety net to reduce overly optimistic failures: if the extracted `guidance_trend` indicates a guidance cut/lowered outlook, or if `momentum` indicates deceleration, any optimistic verdict is overridden to `DOWN` with high confidence. This hybrid design (LLM inference + symbolic gating) enforces high-priority financial heuristics.

### C. Transformations (PCI generators)

Following FinTrust-style consistency testing, we evaluate four statement-level transformations:

- **Negation:** lexical polarity swaps using a small replacement dictionary (e.g., *increase↔decrease*, *growth→decline*). If no trigger word is present, the statement is unchanged.
- **Symmetry (stress test):** reverse the character sequence of the statement as a deliberately strong surface-form perturbation. While not linguistically fluent, it probes sensitivity to superficial form changes.
- **Transitive (entity substitution):** replace the current company name with a top peer company name within the

same sector (via an external sector-to-company mapping) and rewrite first-person pronouns to explicit entity mentions.

- **Additive:** concatenate the statement with the next statement in the dataset order (typically adjacent within the same call context) to test stability under added information.

### D. Decision Consistency Rate (DCR)

For each transformation $T$, let $d_{\text{orig}}$ be the original decision and $d_T$ the transformed decision. We compute **DCR** by restricting to samples where both decisions are non-neutral:

$$\text{DCR}(T) = \frac{1}{n_T} \sum_{i=1}^{n_T} \mathbb{I}[\phi_T(d_{\text{orig}}^{(i)}, d_T^{(i)})] \times 100.$$

For **Negation**, $\phi_T$ expects a decision flip ($d_T \neq d_{\text{orig}}$). For **Symmetry/Transitive/Additive**, $\phi_T$ expects invariance ($d_T = d_{\text{orig}}$). In our dataset, $n_T$ varies by transformation due to `NEUTRAL` filtering (Negation: 104, Symmetry: 99, Transitive: 113, Additive: 101).

### E. Prompt Consistency Index (PCI) for quarter-level diagnostics

We additionally report a quarter-aggregated **PCI** for visualization and temporal diagnostics. We map decisions to numeric values: $v(\texttt{UP}) = +1$, $v(\texttt{DOWN}) = -1$, $v(\texttt{NEUTRAL}) = 0$. For each sample under transformation $T$, we assign a binary consistency score $s_T \in \{0, 100\}$. For **Symmetry/Transitive/Additive**, $s_T = 100$ iff $v(d_T) = v(d_{\text{orig}})$. For **Negation**, $s_T = 100$ iff $v(d_T) \cdot v(d_{\text{orig}}) = -1$ (UP↔DOWN), and we treat $v(d_T) = v(d_{\text{orig}}) = 0$ (NEUTRAL↔NEUTRAL) as consistent. We then average $s_T$ within each quarter $q$:

$$\text{PCI}_q(T) = \frac{1}{|S_{q,T}|} \sum_{i \in S_{q,T}} s_T^{(i)}.$$

Compared to DCR, PCI includes abstentions and is used as a diagnostic visualization score rather than the primary model-comparison metric.

### F. Visualization

We summarize model behavior with standard visual diagnostics: (i) quarter-level PCI radar and heatmap / parallel-coordinates plots to reveal temporal and transformation-wise stability trends, and (ii) DCR as the primary strict consistency score for model-to-model comparison under each transformation.

## V. Experiments

### A. Task Definition and Label Construction

We study **3-day stock direction prediction** (UP vs. DOWN) from earnings-call statements. Let $t_0$ denote the **earnings-call day close** if the call occurs on a trading day; otherwise, $t_0$ is set to the **next trading day** (close price). We define the 3-trading-day return as:

$$r = \frac{P_{t_0+3} - P_{t_0}}{P_{t_0}},$$

where $P_{t_0}$ and $P_{t_0+3}$ are close prices. The ground-truth label is **UP** if $r > 0$, and **DOWN** otherwise (no neutral class).

### B. Dataset

Our dataset contains **486** labeled samples spanning **8 quarters (2024 Q1–2025 Q4)** and **12 companies** across four sectors (Technology, Financials, Retail/Consumer, and Healthcare). The label distribution is near-balanced (**UP: 252**, **DOWN: 234**).

### C. Model: Trinity (GPT-4o + Prompt/Rule Augmentation)

We implement **Trinity**, a GPT-4o-based prompting system with three key components: **(i) Hybrid Context Retrieval** that combines a tone-preserving anchor with keyword-focused evidence to retain both global narrative and local signals, **(ii) Structured Reasoning Template** that enforces a fixed pipeline (*Extraction* → *Debate* → *Verdict*), and **(iii) Python-side Hard Rules** that override predictions under high-risk patterns (e.g., guidance cuts and momentum deceleration). We use deterministic decoding (temperature $= 0$) and require JSON-formatted outputs.

### D. Baselines

We include the baseline model families reported in *Measuring Consistency in Text-based Financial Forecasting Models* [4], including EVENT, HTML, MRDM, and XGB, as **historical references**. Due to time and computational constraints, we **do not re-train or re-evaluate** these baselines on our 2024–2025 dataset in this report. As a result, baseline numbers shown in the following comparisons should be interpreted as **contextual anchors** rather than a fully controlled, same-distribution benchmark.

### E. Evaluation Metrics

We report **Accuracy**, **Macro-F1**, and **Matthews Correlation Coefficient (MCC)** for direction prediction.

### F. Consistency Metrics: DCR and PCI

Following [4], we evaluate semantic consistency under four transformation types: **Negation**, **Symmetry**, **Transitive**, and **Additive**.

*a) Directional Consistency Rate (DCR).:* Let $\hat{y}$ be the predicted direction for the original input and $\hat{y}^{(v)}$ for a transformed variant $v$. For **Negation**, consistency is satisfied when $\hat{y}^{(\text{neg})} \neq \hat{y}$ (label flip). For **Symmetry/Transitive/Additive**, consistency requires invariance $\hat{y}^{(v)} = \hat{y}$. We compute DCR as the percentage of consistent cases among evaluated variants.

*b) Quarter-level PCI Profiles.:* We additionally summarize consistency *per quarter* as PCI profiles (heatmap/radar/parallel coordinates), to visualize temporal stability and highlight which transformation types are systematically difficult. DCR emphasizes directional logic (flip vs. invariance), while PCI serves as an aggregated diagnostic view across quarters.

## VI. Results

### A. Predictive Performance (3 Trading Days)

Table II reports direction prediction performance under the 3-trading-day horizon. Trinity achieves the highest Accuracy and MCC among all compared methods, indicating stronger directional signal capture on the evaluated set.

| Model | Accuracy | Macro-F1 | MCC |
|---|---|---|---|
| EVENT | 0.442 | 0.608 | 0.019 |
| HTML | 0.465 | 0.569 | -0.024 |
| MRDM | 0.447 | 0.598 | -0.032 |
| XGB | 0.462 | 0.456 | -0.076 |
| Trinity (GPT-4o) | 0.585 | 0.584 | 0.176 |

TABLE II
3-TRADING-DAY DIRECTION PREDICTION PERFORMANCE (UP VS. DOWN).

*Disclaimer.* Some baseline results are directly cited from [4] and may come from an earlier data period. Therefore, cross-table comparisons may be affected by **distribution shift** between 2024–2025 disclosures and prior datasets. We treat these baseline numbers as **historical reference points** to contextualize the evolution from traditional models to modern LLM prompting, rather than as a strictly controlled, same-distribution benchmark.

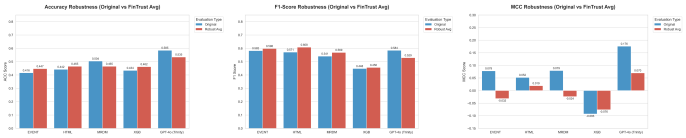


Fig. 1. Robustness summary (Original vs. FinTrust Avg) across Accuracy, Macro-F1, and MCC.

**How to read Fig. 1.** Blue bars show each method's performance on the *original* statements. Red bars (FinTrust Avg) report the corresponding metric averaged over FinTrust-style transformed variants (Negation, Symmetry, Additive, Transitive). The key signal is the **within-method gap** between blue and red: larger drops indicate higher sensitivity to wording-level perturbations and weaker robustness. Because some baseline numbers may be drawn from prior work, we interpret this figure primarily through **relative degradation** (Original $\rightarrow$ FinTrust Avg) rather than absolute cross-method score gaps.

### B. Consistency Under Semantic Transformations (DCR)

Figure 3 and Table III summarize directional consistency under semantic transformations. Trinity is highly consistent under **Symmetry**, **Transitive**, and **Additive** (all $\approx$ 98–99%), but exhibits a near-zero flip rate under **Negation**, indicating that simple polarity flips in a single statement do not reliably change its directional verdict.

### C. Quarter-level PCI Profiles (Diagnostic View)

Figure 4 and Figure 5 visualize quarter-level PCI profiles. Across quarters, **Transitive** and **Additive** remain consistently high, while **Negation** is the most volatile dimension, with noticeable drops in certain quarters (e.g., 2025 Q1).

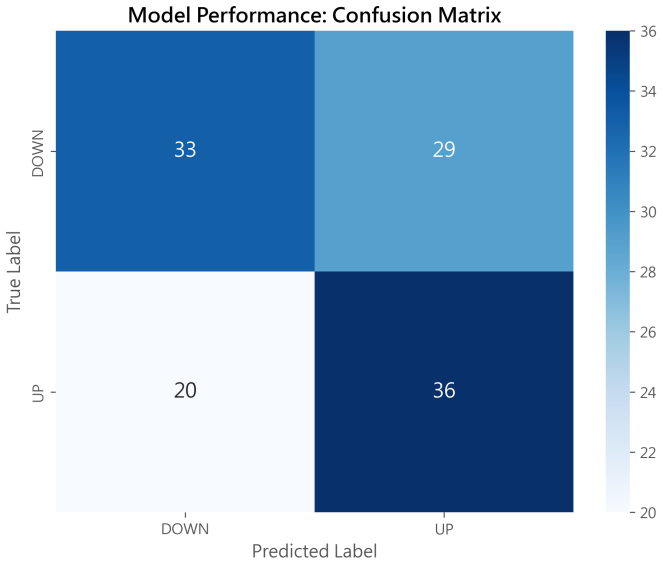


Fig. 2. Confusion matrix of Trinity on instances where it outputs a directional decision (UP/DOWN).

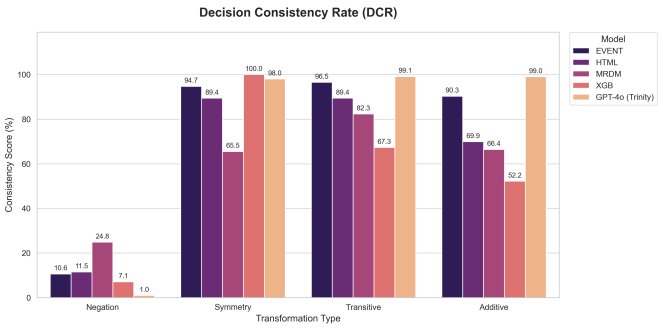


Fig. 3. **DCR** across semantic transformations. Negation expects label flipping; Symmetry/Transitive/Additive expect invariance.

| Model | Negation | Symmetry | Transitive | Additive |
|---|---|---|---|---|
| EVENT | 10.6 | 94.7 | 96.5 | 90.3 |
| HTML | 11.5 | 89.4 | 89.4 | 69.9 |
| MRDM | 24.8 | 65.5 | 82.3 | 66.4 |
| XGB | 7.1 | 95.8 | 67.3 | 52.2 |
| Trinity (GPT-4o) | 1.0 | 98.0 | 99.1 | 99.0 |

TABLE III
DIRECTIONAL CONSISTENCY RATE (DCR, %) BY TRANSFORMATION TYPE.

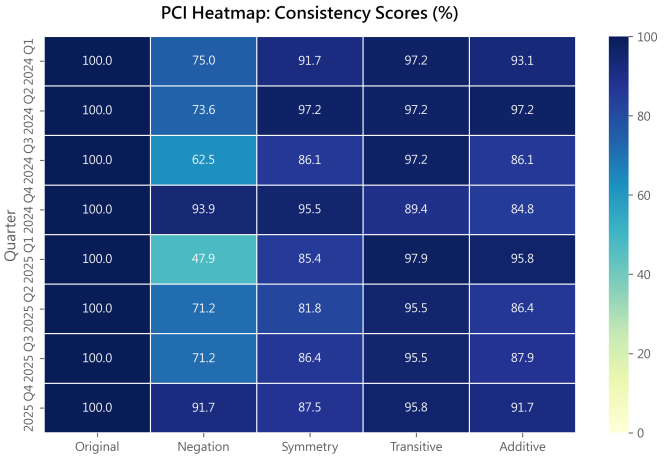


Fig. 4. Quarter-level PCI heatmap across transformation types.

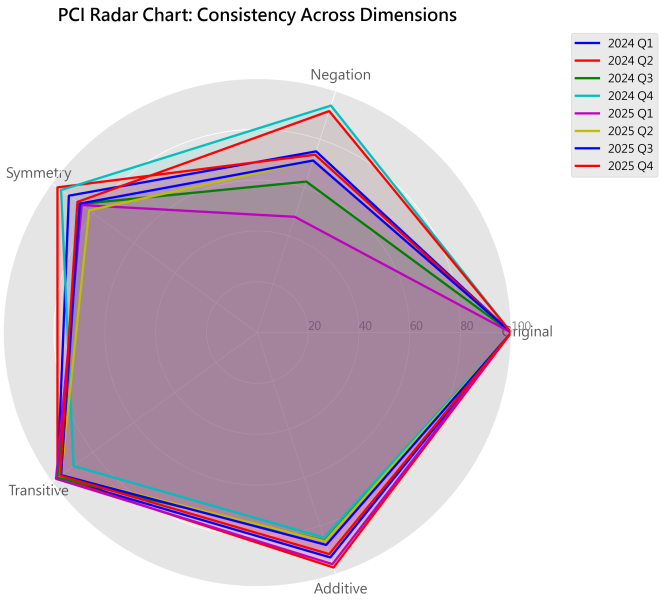


Fig. 5. Quarter-level PCI radar chart across transformation types.

## VII. Future Work

In future work, we will (i) extend our evaluation to 7/15/30-day trading horizons to better stress-test temporal robustness, (ii) enforce an explicit *reasoning-consistency* (contradiction) check in Phase 2 (Debate) to reduce unstable verdicts, and (iii) improve overall accuracy (target $\geq$60%) and strengthen *negation* consistency—a shared weakness observed in both prior work [4] and our study. To support these goals, we will scale the input corpus via web scraping to cover more industries and companies.

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